

Java Programming

Overview

Java is a general-purpose, object-oriented programming language designed around the idea of writing code once and running it on many platforms. Java source code is compiled into bytecode, which can run on the Java Virtual Machine (JVM).

Variables and Types

Java is statically typed, meaning variables have declared types. Common primitive types include int, double, boolean, and char. Java also provides reference types such as String, arrays, and user-defined classes.

Object-Oriented Programming

Java heavily uses object-oriented programming. Classes define the structure and behavior of objects, while inheritance, encapsulation, abstraction, and polymorphism allow programmers to organize larger applications.

Collections

Java provides collection classes such as ArrayList, HashMap, HashSet, and LinkedList. These structures are useful for storing and retrieving groups of objects efficiently.

Applications

Java is widely used for backend services, enterprise software, Android development, distributed systems, and large-scale applications.